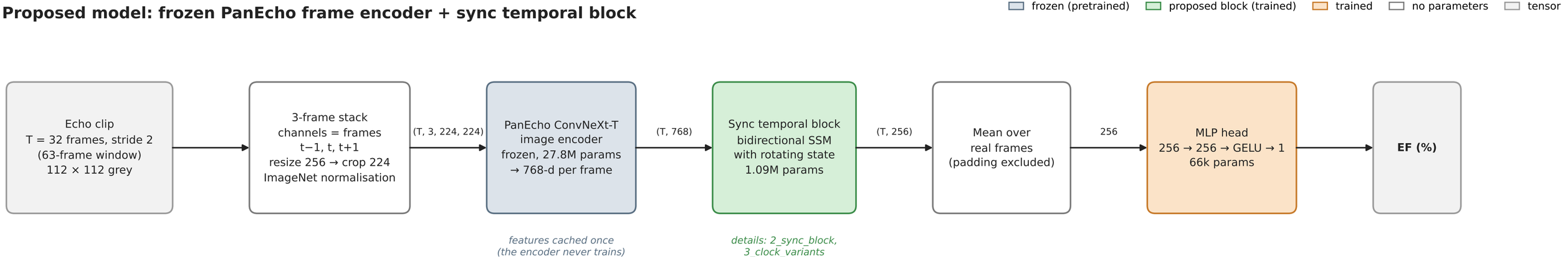
Proposed model: frozen PanEcho frame encoder + sync temporal block



Training: MSE on standardised EF, AdamW lr 1e-3 (cosine, 5% warm-up), batch 16, 40 epochs; weight EMA 0.999 for validation, checkpoint selection and evaluation. Evaluation: mean EF over 4 evenly spaced clips per video.

Frame encoder of the final model: PanEcho ConvNeXt-Tiny (frozen, 27,820,128 parameters)

frozen (pretrained) | no parameters | tensor

video frames
112 × 112 RGB

→ resize 256 × 256
(cv2 INTER_AREA)

→ centre crop
224 × 224

→ 3-frame stack: channel j
= frame t−1+j (clamped
at the video ends)

→ ÷ 255, ImageNet mean/std
normalisation

→ x_t : (3, 224, 224)
one image per frame

x_t →

stem
Conv 4×4, stride 4
3 → 96 + LayerNorm
4,896

→ stage 1
3 ConvNeXt blocks
96 ch, 56 × 56
237,888

→ down
LN + Conv 2×2 s2
96 → 192
74,112

→ stage 2
3 blocks
192 ch, 28 × 28
918,144

→ down
192 → 384
295,680

→ stage 3
9 blocks
384 ch, 14 × 14
10,817,280

→ down
384 → 768
1,181,184

→ stage 4
3 blocks
768 ch, 7 × 7
14,289,408

→ global avg pool
+ LayerNorm
1,536

→ **f_t : 768**

one ConvNeXt block (CNBlock), used 18 times:

input
(C, h, w)

→ depthwise Conv 7×7
(padding 3)

→ LayerNorm
(channels)

→ Linear C → 4C

→ GELU

→ Linear 4C → C

→ layer scale γ
(per channel)

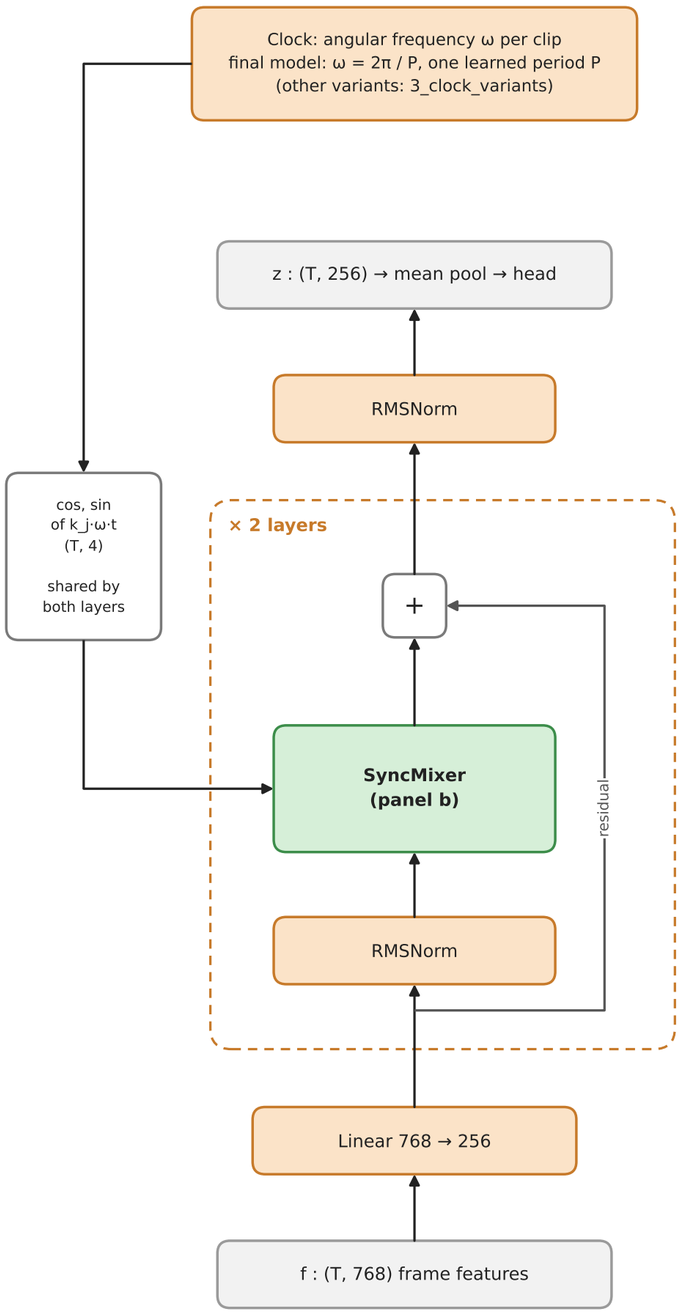
→ + input
(residual)

Weights: PanEcho release (Holste et al., JAMA 2025; trained on >1M Yale echocardiography videos; EchoNet-Dynamic not used), loaded as tensors only, CC BY-NC-SA 4.0. Never trained here: run once in bf16 over every frame of every video, features stored as float16 (10,030 videos, 1.77M frames). Stochastic depth is inactive (eval mode).

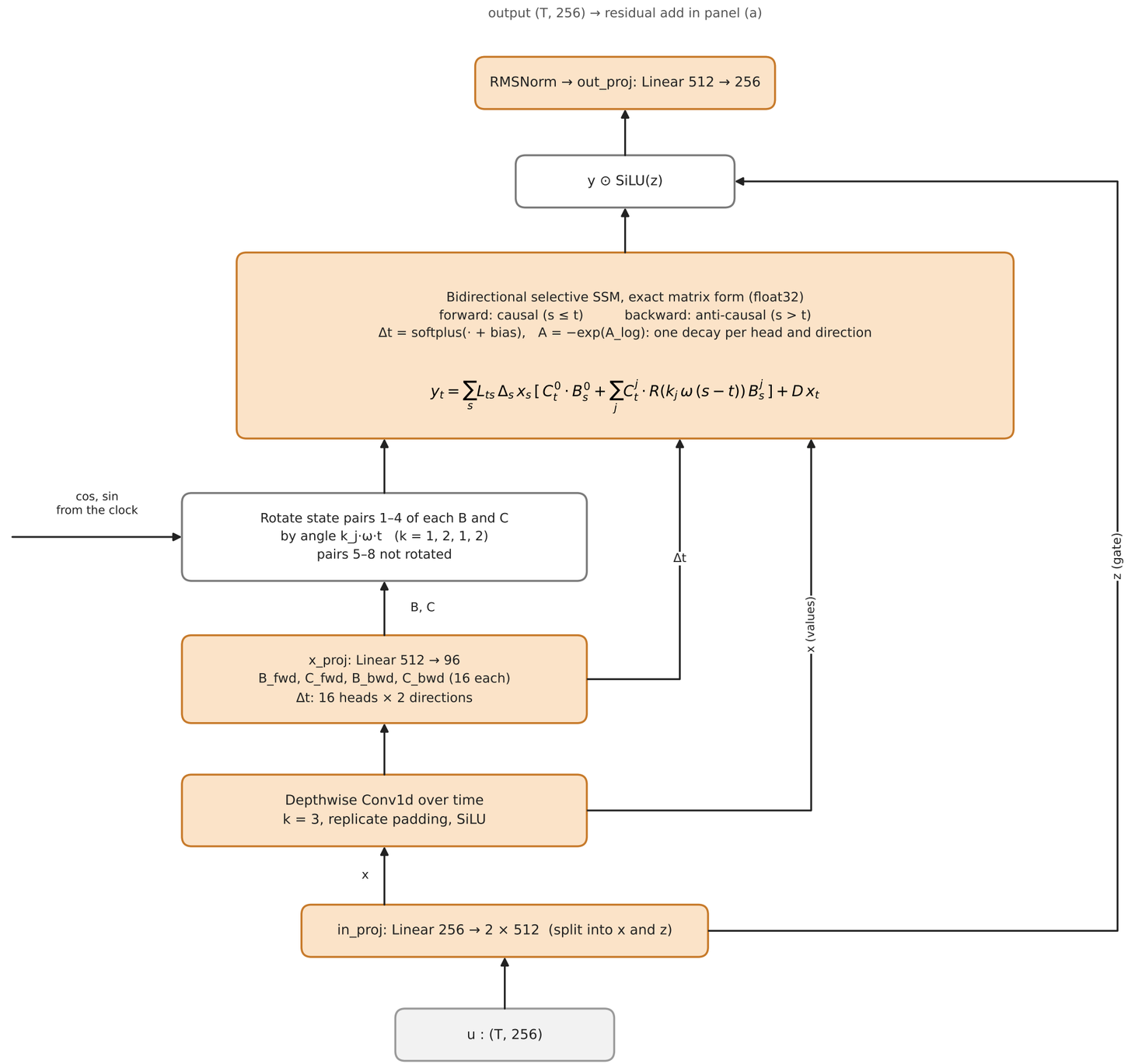
Sync temporal block (proposed)

trained · proposed block (trained) · no parameters · tensor

(a) SyncTemporal — 2 layers



(b) SyncMixer — one layer: d_model 256, d_inner 512, 16 heads × 32 per direction, state N = 16



Final model: every trainable tensor, hyperparameters and training recipe

Trainable parameters (counted from the model)

input projection	Linear 768 → 256 (weight + bias)	196,864	1×
layer ℓ = 1, 2: pre-norm	RMSNorm(256)	256	2×
mixer in_proj	Linear 256 → 1024 (no bias): x (512) and gate z (512)	262,144	2×
mixer conv	depthwise Conv1d over time, k = 3, 512 channels (+ bias), replicate padding	2,048	2×
mixer x_proj	Linear 512 → 96 (no bias): B_fwd, C_fwd, B_bwd, C_bwd (16 each) + Δt (32)	49,152	2×
dt_bias	32 (16 heads × 2 directions); Δt = softplus(x_proj + dt_bias)	32	2×
A_log	32; A = −exp(A_log), one decay per head and direction	32	2×
D	16; skip term D · x	16	2×
output norm	RMSNorm(512) after y ⊙ SiLU(z)	512	2×
mixer out_proj	Linear 512 → 256 (no bias); + residual	131,072	2×
final norm	RMSNorm(256)	256	1×
clock	log P: one learned period, ω = 2π / P (init 24 frames)	1	1×
head	Linear 256 → 256, GELU, Linear 256 → 1	66,049	1×
total trainable	temporal block 1,087,649 + head 66,049	**1,153,698**	

Forward pass for one clip

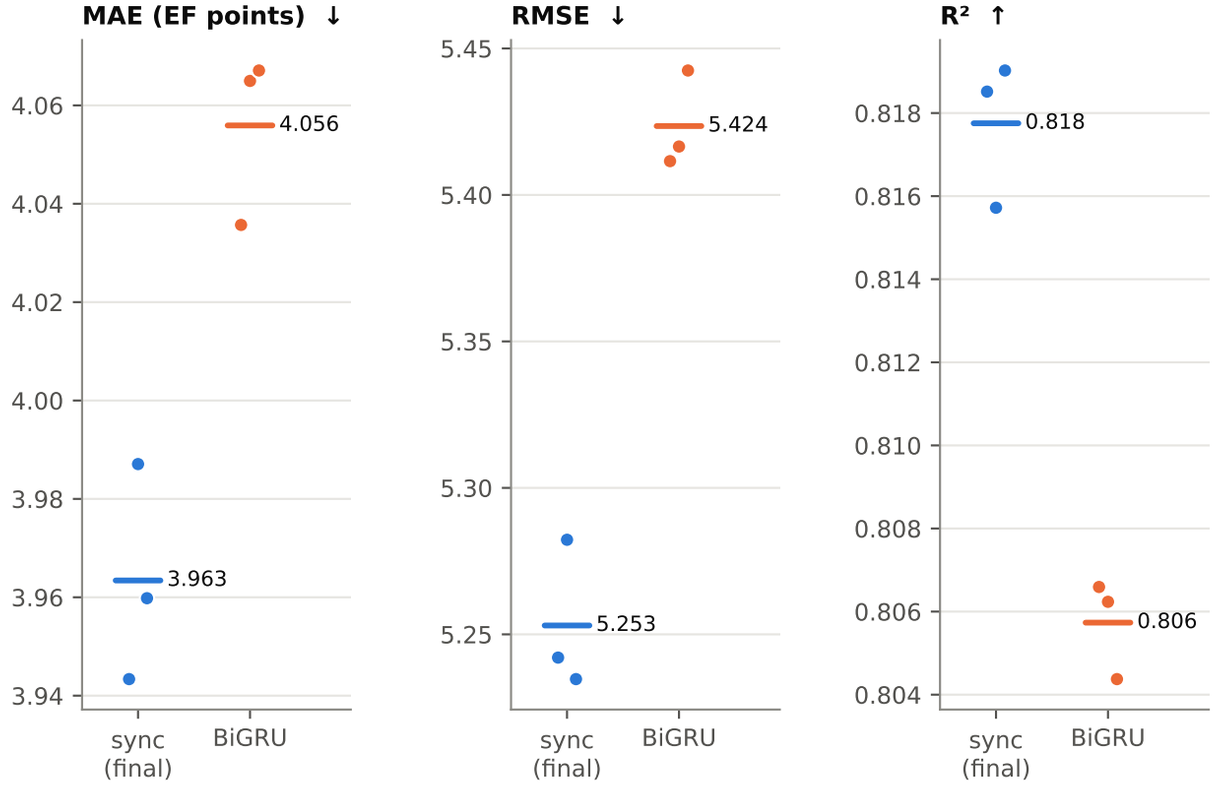
f (T, 768) → Linear → x (T, 256)
ω = 2π / P; cos, sin of k_j · ω · t for t = 0 … T−1, j = 1 … 4 (computed once, shared by both layers)
for each layer: u = RMSNorm(x); [x′, z] = in_proj(u); x′ = SiLU(Conv1d(x′)); B, C, Δt = x_proj(x′)
  rotate pairs 1–4 of B and C by angle k_j ω t; y = forward SSM (s ≤ t) + backward SSM (s > t) + D·x′, as one T × T matrix per head
  x = x + out_proj(RMSNorm(y ⊙ SiLU(z)))
z = RMSNorm(x) → mean over real (non-padding) frames → head → standardised EF → EF = 12.3·ŷ + 55.75 (TRAIN SD / mean)

Hyperparameters
d_model 256, expand 2 → d_inner 512
16 heads × head dim 32 per direction
state size N = 16 per direction
rotating state pairs R = 4 of 8,
  harmonics k = (1, 2, 1, 2)
Δt init log-uniform [1e-3, 1e-1]
A init uniform [1, 16]
learned period P after training:
  26.2 / 24.9 / 25.5 sampled frames
  (seeds 0 / 1 / 2) ≈ 1 s ≈ 1 beat

Training
frozen features, T = 32 frames, stride 2,
  random clip start, loop padding
MSE on standardised EF
AdamW, lr 1e-3, weight decay 1e-4
cosine schedule, 5% warm-up
batch 16, 40 epochs, bf16 autocast
weight EMA 0.999 (used for validation,
  selection and evaluation)
best epoch by single-clip VAL MAE

Evaluation
EF = mean over 4 evenly spaced clips
  per video; mean over real frames

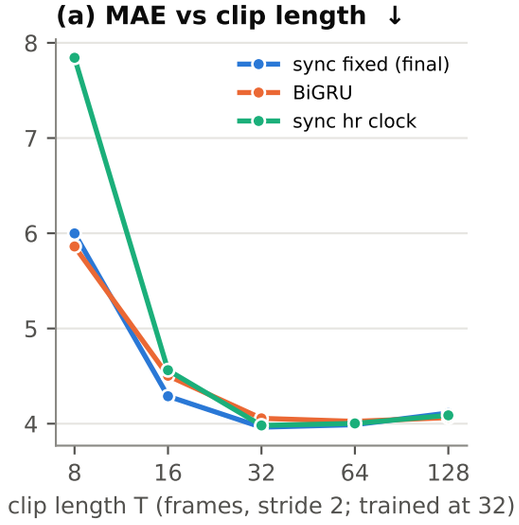
Final model vs BiGRU — validation set (frozen PanEcho features, weight EMA, 3 seeds, 4 clips per video)

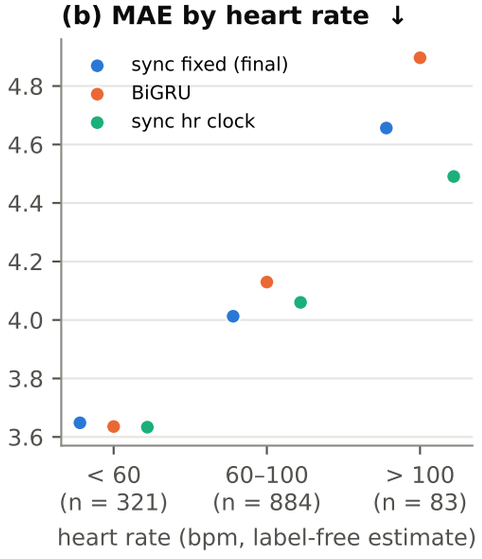


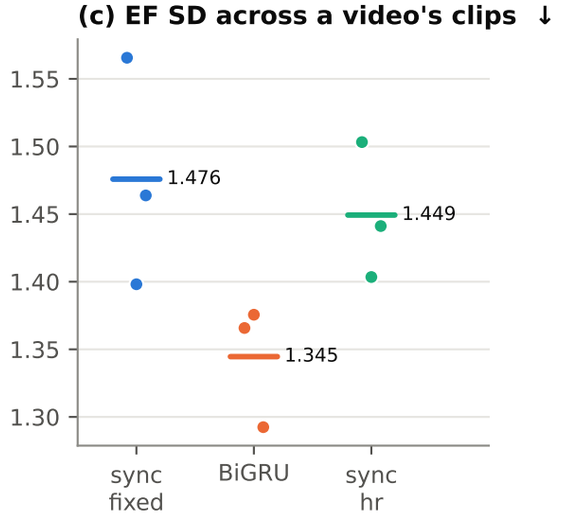
metric	sync	BiGRU	better
MAE (EF points)	3.963	4.056	sync
RMSE	5.253	5.424	sync
R^2	0.818	0.806	sync
Pearson r	0.905	0.900	sync
CCC	0.899	0.890	sync
|bias|	0.188	0.739	sync
within 5 points (%)	71.1	70.1	sync
within 10 points (%)	93.7	93.0	sync
AUROC, EF < 40%	0.984	0.983	sync
AUROC, EF < 50%	0.960	0.959	sync

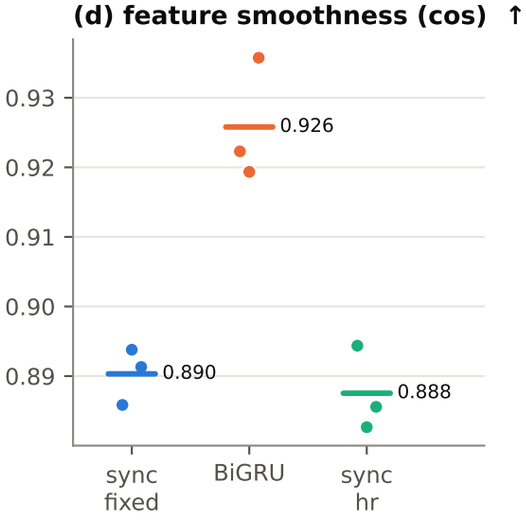
Means of 3 seeds. Paired over 1288 validation videos (seed-averaged errors):
ΔMAE (sync − BiGRU) = -0.093 [95% CI -0.164, -0.020], p = 0.011

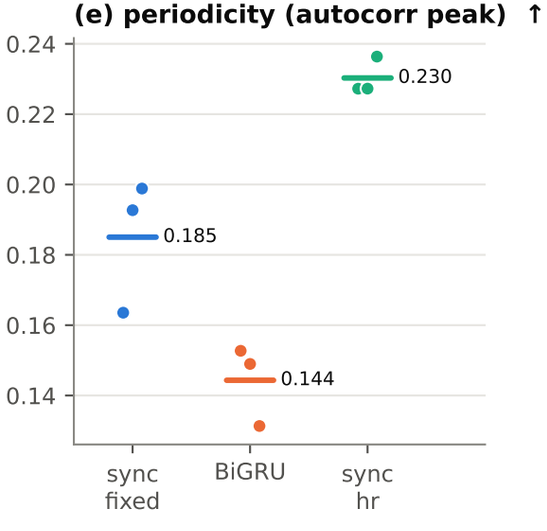
Sync block vs BiGRU on every evaluation metric — validation set (mean of 3 seeds; dots = seeds)

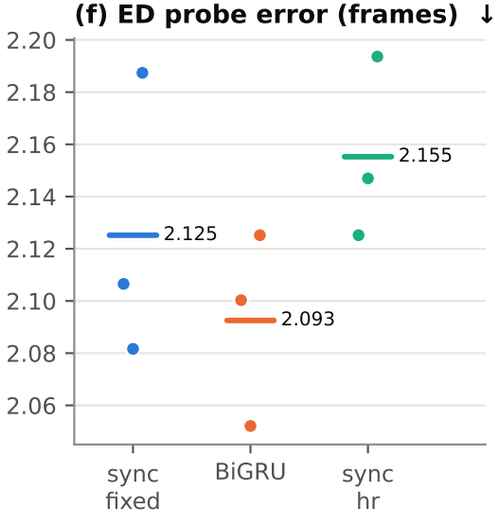


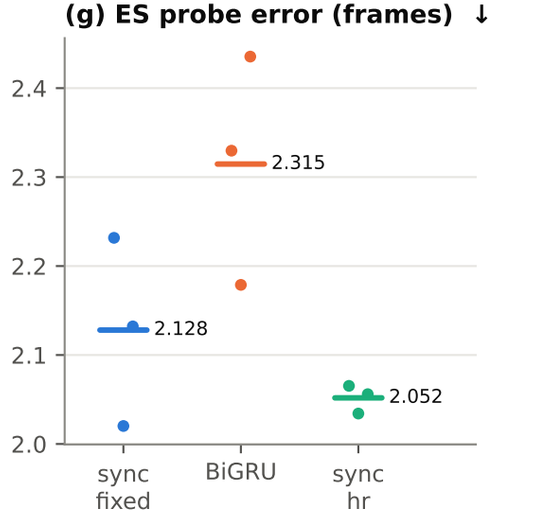










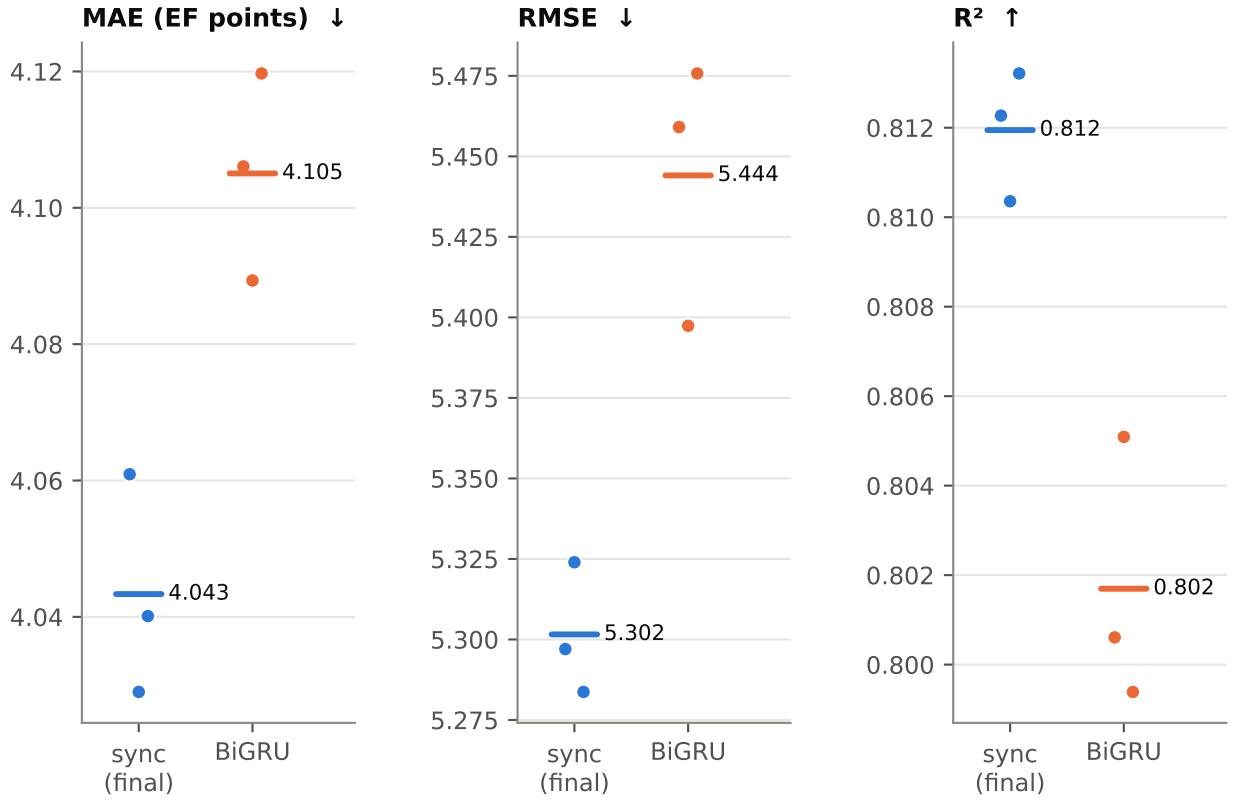


(h) efficiency of the head, T = 32 ↓

model	params	GFLOP/clip	ms/clip b16	train ms/step
sync fixed	1.15M	0.072	0.082	10.6
BiGRU	1.15M	0.069	0.040	3.6
sync hr	1.15M	0.087	0.131	11.3

frozen encoder (shared) excluded; RTX 3050, CUDA graphs, batch 16

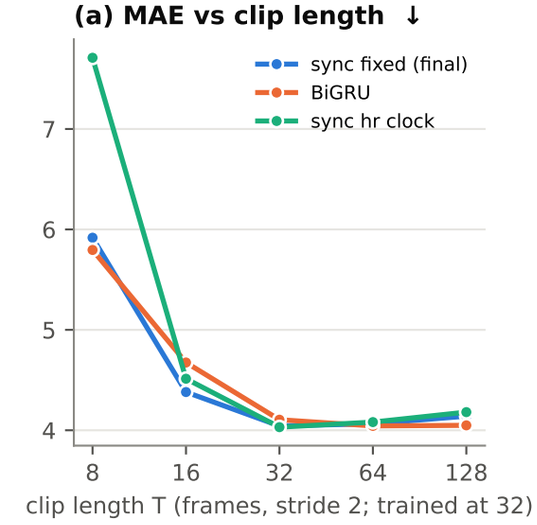
Final model vs BiGRU — test set (frozen PanEcho features, weight EMA, 3 seeds, 4 clips per video)

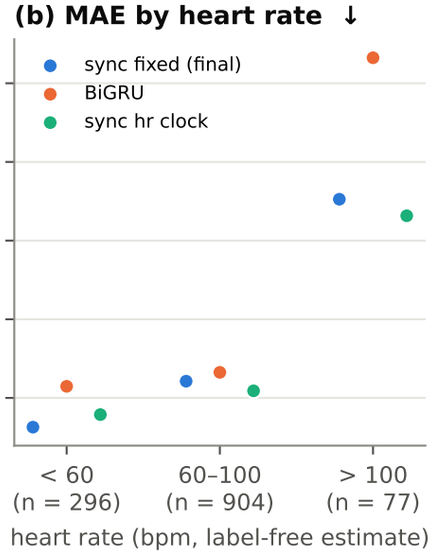


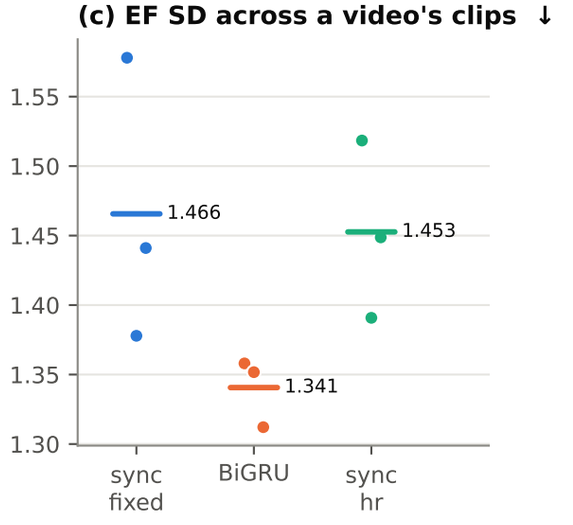
metric	sync	BiGRU	better
MAE (EF points)	4.043	4.105	sync
RMSE	5.302	5.444	sync
R^2	0.812	0.802	sync
Pearson r	0.902	0.900	sync
CCC	0.894	0.886	sync
|bias|	0.444	0.995	sync
within 5 points (%)	70.2	69.4	sync
within 10 points (%)	93.2	93.0	sync
AUROC, EF < 40%	0.983	0.981	sync
AUROC, EF < 50%	0.966	0.967	BiGRU

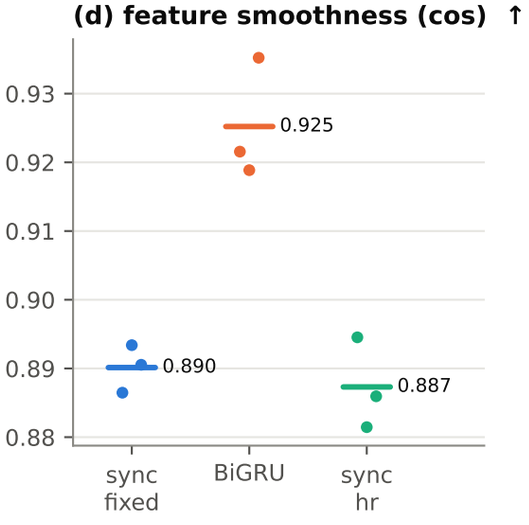
Means of 3 seeds. Paired over 1277 test videos (seed-averaged errors):
ΔMAE (sync − BiGRU) = -0.062 [95% CI -0.132, +0.011], p = 0.093

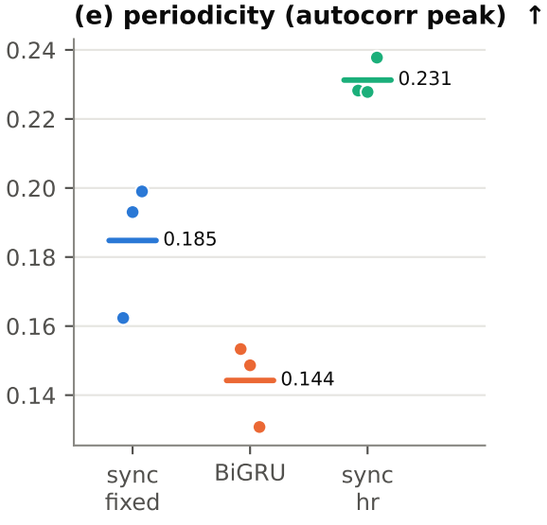
Sync block vs BiGRU on every evaluation metric — test set (mean of 3 seeds; dots = seeds)

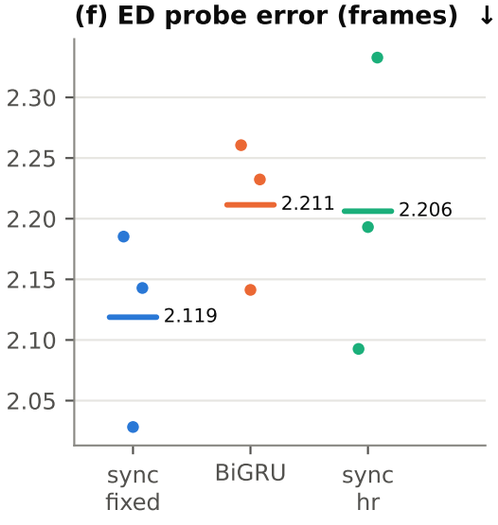


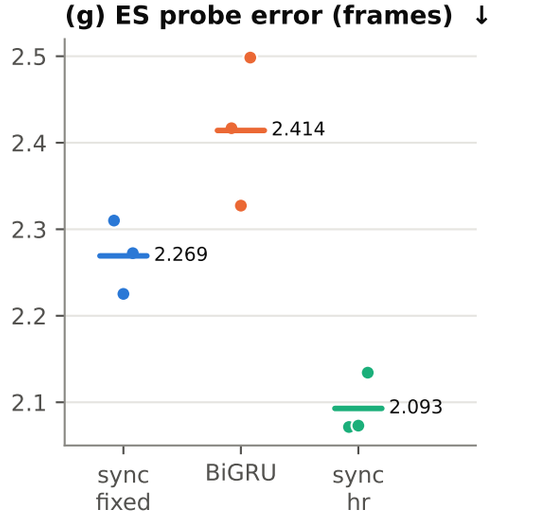












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